TETHER: Tethered Causal Temporal Refinement for Frozen Surgical Stereo

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***Abstract*— Dense stereo is attractive for robotic surgery because it provides metric geometry without active illumination, yet modern stereo estimators are typically applied independently to each frame and can therefore fluctuate under specularities, weak texture, tissue deformation, tool occlusions, and camera motion. We ask whether temporal information can improve frozen stereo predictions without modifying the stereo estimator, accessing its internal representation, or using future frames.**

We present TETHER, a causal plug-and-play temporal refiner for frozen stereo. The name reflects the constraint that defines it: every output is tethered between the current stereo estimate and aligned temporal memory, and the recurrent state is periodically re-tethered to independently predicted raw geometry. At each frame, preceding disparity is motion-aligned to the current view using frozen SEA-RAFT. A shared 142-channel external evidence representation feeds a 177,338-parameter CODD-inspired reset-and-fusion head that predicts a bounded interpolation between the current stereo estimate and aligned temporal memory. Corrected recurrence is explicitly bounded and periodically re-anchored to independently predicted raw geometry.

A controlled development-set closure shows that the learned policy is not explained by temporal smoothing: on SCARED-C D2, canonical $H = 4$ reduces mean EPE by 4.05%, compared with only 0.59% for the best matched fixed/EMA blend, while direct flow-confidence weighting and ungated memory propagation degrade accuracy. The raw-versus-memory oracle exposes a 12.30% improvement ceiling, of which TETHER recovers 32.9%. Continuous corrected-state recurrence increases EPE by 62.69%, supporting bounded memory.

One frozen $H = 4$ checkpoint subsequently improves all five evaluated stereo estimators on held-out SCARED-C D7, including CREStereo and Fast-FoundationStereo, which were excluded from temporal-head training. Mean backbone EPE decreases from 0.4921 to 0.4696, while Bad1 and disparity RMSE also decrease for every backbone. Zero-shot evaluation on five DRENDS recordings ($20{,}929$ frames) with three estimators, two of them unseen in training, improves disparity EPE, Bad3 and metric depth in all 60 recording–backbone–metric cells, completing the transfer matrix across both the backbone and the domain shift. We further show that confining the fused output to the support the module already declares is load-bearing rather than cosmetic: without it, zero-padded out-of-bounds memory turns rare pixels into invalid predictions and produces the metric-depth tail previously attributed to the external domain.

We additionally introduce a unified refinement-centric evaluation framework that jointly measures spatial geometry, multi-horizon temporal fidelity, introduced and recovered error, frame-level tails, and selective risk under fixed prediction-independent support.

I. Introduction

Metric three-dimensional perception underpins camera navigation, tool–tissue distance estimation, augmented reality and deformation analysis in robot-assisted surgery, and stereo endoscopy is attractive because depth is recovered passively from images the platform already provides.

It is also hard. Endoscopic scenes carry weak or repetitive texture, non-Lambertian tissue, specular highlights, blood, smoke, deformable anatomy, rapid viewpoint change and frequent tool occlusion [20], [21]. Modern estimators [12]–[17] give increasingly strong frame-wise disparity, but nothing in a frame-wise model requires its predictions to cohere over time. Video stereo methods address this by building temporal reasoning into the model [1]–[4], at the price of assuming a temporal architecture, model-specific representations, future observations, or co-design with the estimator.

We study a different deployment setting:

> *Can one causal temporal module be trained once and attached unchanged to heterogeneous frozen stereo estimators?*

This formulation matters for robotic systems because the stereo backbone and temporal logic can evolve independently. A new stereo estimator can replace the previous one without requiring the temporal module to access its cost volume, hidden state, confidence head, or training pipeline.

Our starting point is the reset-and-fusion principle introduced by CODD [1]. We do not claim reset/fusion itself as novel. Instead, we retain its useful inductive bias: the temporal system should decide whether motion-aligned historical geometry should influence the current stereo estimate and constrain the correction to a convex interpolation between the two.

We redesign this principle for an external black-box interface. The integrated motion pathway is replaced by frozen SEA-RAFT; stereo-backbone-specific representations are replaced by a shared external evidence space; and the same learned temporal head is evaluated unchanged across heterogeneous stereo architectures.

The resulting system has a deliberately constrained structure. The current stereo estimate remains an anchor. Temporal memory is geometrically aligned before use. The learned head controls the amount of intervention, but cannot produce a value outside the interval spanned by the current and temporal candidates. Finally, corrected recurrence is periodically reset to raw stereo geometry.

These restrictions raise an important question: does the learned head actually provide value, or would simple temporal averaging give the same improvement?

We answer this experimentally under a frozen development protocol. Matched fixed fusion, EMA, warped-memory propagation, and direct forward–backward confidence weighting recover only a small fraction of TETHER's gain. In contrast, a ground-truth oracle shows that the aligned memory contains substantially more useful information than the current policy exploits. This indicates that the core problem is not obtaining temporal evidence, but identifying *where and how much* of it should be used.

A second question is how such intervention should be evaluated. Temporal smoothness is not equivalent to temporal correctness: a stale prediction can be smooth while being geometrically wrong. Likewise, lower mean EPE does not reveal whether refinement introduces new bad pixels or rare catastrophic frames. We therefore evaluate refinement relative to the frozen upstream prediction and explicitly separate geometry, temporal fidelity, benefit, harm, and tail behavior.

Our contributions are three.

We formulate surgical temporal stereo refinement as a *causal plug-and-play adaptation problem* over frozen, heterogeneous estimators, needing neither future frames nor backbone internals, and instantiate it as TETHER: a 177,338-parameter reset-and-fusion head over causal SEA-RAFT alignment, a shared 142-channel external evidence space, support-confined convex fusion and bounded corrected-state recurrence.

We show through a matched-baseline closure that the gain is not temporal smoothing — 4.05% against 0.59% for the best fixed blend under an identical state machine — and that one frozen checkpoint transfers along both available axes: to two estimators excluded from training on held-out D7, and to five recordings of an unseen domain, giving the first measurement of the two shifts combined.

We introduce a refinement-centric evaluation protocol under fixed, prediction-independent support that measures introduced as well as recovered error, and we use it to bound our own claims: degenerate history-replay controls win the no-reference consistency metric outright, and the head's internal weight inverts its relation to harm between development and held-out data.

II. Related Work

a) Deep stereo matching.: Stereo has moved from learned matching costs to iterative recurrent estimation and broad-domain zero-shot models: RAFT-Stereo [12], CREStereo [13], Stereo Anywhere [14], S^2M^2 [15] and foundation models [16], [17]. Our work is orthogonal: we neither modify the estimator nor assume an architecture.

b) Temporal stereo and depth.: CODD couples stereo, motion and fusion online and introduced the reset/fusion mechanism we build on [1]. DynamicStereo applies spatio-temporal attention to stereo video [2], PPMStereo learns pick-and-play memory inside an integrated architecture [4], and TC-Stereo propagates a temporal state causally [6]. All four are co-designed with the estimator: they replace it, retrain it, or read its hidden state. Match Stereo Videos exposes BiDAStabilizer as a plug-in [3], but propagates bidirectionally and therefore consumes future frames. NVDS stabilises monocular relative depth [18], a different output space. A separate line achieves temporal robustness by adapting the stereo network's weights online [10], which is the opposite of the constraint we adopt.

c) The comparison that does not exist.: Our setting — causal, black-box, frozen weights — admits, to our knowledge, one published method: StereoDiffusion refines flow-warped disparity from an off-the-shelf estimator with a diffusion prior and evaluates on SCARED [7]. Its repository contains no implementation, so we could not run it. Of the remaining candidates, the black-box spatial refiner of [8] and the model-agnostic video-consistency network of [9] publish code whose weights are no longer retrievable, and TC-Stereo's temporal path requires per-frame camera pose and warps by a rigid transform, an assumption deformable tissue violates and SCARED-C cannot supply. We therefore compare against the one method that is both runnable and published, BiDAStabilizer, and state plainly that it sees the future while we do not.

d) Evaluating temporal refinement.: Stereo benchmarks report frame-wise geometry and video methods add consistency measures, but neither characterises an intervention: a prediction can grow smoother without growing more correct, and a lower mean error can hide newly created failures. Refinement is comparative, so recovered and introduced error must both be measured, which connects to selective prediction [19]. Our contribution is not a new metric but a common protocol under fixed, prediction-independent support.

e) Surgical stereo geometry.: SCARED [20], SERV-CT [21], D4D [23] and DRENDS [24] supply the endoscopic stereo, CT-derived and structured-light geometry used here; Sec. IV states what each can and cannot support.

III. Method

A. External Causal Interface

A frozen estimator S produces $d_t^{\text{raw}}, M_t^{\text{raw}} = S(I_t^L, I_t^R)$, positive-left disparity and its validity mask. The temporal module never reads the internal representation of S; it consumes only externally observable quantities — RGB, disparity, validity, optical flow and evidence derived from them. At time t the cache holds the preceding left image, the preceding raw disparity and validity, the previous fused state and its age; nothing from $t+1$ onwards exists. The map to learn is therefore $d_t^{\text{fused}} = T_\theta(d_t^{\text{raw}}, m_{t-1}, I_t^L, I_t^R, I_{t-1}^L)$.

B. Causal Motion Alignment

Frozen SEA-RAFT [5] estimates both directions between the two already observed left images, $F_{t\to t-1} = F(I_t^L, I_{t-1}^L)$ and $F_{t-1\to t} = F(I_{t-1}^L, I_t^L)$; the second is used only to measure reliability. Bidirectional *flow estimation* between two observed frames is not bidirectional temporal inference.

Previous geometry is pulled onto the current grid, $\widetilde{m}_t(x) = m_{t-1}(x + F_{t\to t-1}(x))$, by bilinear sampling with zero padding and `align_corners=True`. Warp support S_t records whether the source coordinate lies inside the preceding image, and historical validity is sampled on the same grid. Pulling the reverse flow the same way, $\widetilde{F}_{t-1\to t} =$

$\mathcal{W}(F_{t-1\to t}, F_{t\to t-1})$, gives the forward–backward residual $e_t^{FB} = \|F_{t\to t-1} + \widetilde{F}_{t-1\to t}\|_2$ and the reliability cue

$$C_t^{FB} = S_t \exp\left(-\frac{e_t^{FB}}{0.5 + 0.01\left(\|F_{t\to t-1}\|_2 + \|\widetilde{F}_{t-1\to t}\|_2\right)}\right). \tag{1}$$

The zero padding in this warp is not a detail: it is what makes $\widetilde{m}_t$ exactly zero outside the source image, and Section V-I shows what happens when that value is allowed into the output.

C. Shared External Evidence

All stereo estimators are mapped into one common evidence representation:

$$C_t = \Phi\left(d_t^{\text{raw}}, \widetilde{m}_t, I_t^L, I_t^R, I_{t-1}^L, F_{t\to t-1}, C_t^{FB}, S_t, \widetilde{M}_t\right). \tag{2}$$

The head consumes a single 142-channel tensor built at two resolutions. At quarter resolution, 129 channels carry matching and frozen appearance information: matching curves around raw and memory disparity, disparity and appearance self- and cross-correlations, sampled matching-support cues, and 64 frozen ResNet-18 layer-1 channels [11]. These are upsampled to the cache grid and concatenated with 13 cache-grid channels holding current disparity, aligned memory, signed and absolute disagreement, flow components and magnitude, forward–backward confidence, warp support, aligned validity and RGB evidence.

Normalisation is fixed before training: disparities clipped to $[0, 64]$ and scaled by 64, signed residuals clipped to $[-16, 16]$ and scaled by 16, flow components on $[-32, 32]$ and magnitude on $[0, 32]$, and ImageNet-normalised RGB for the frozen features. No ground truth, backbone identity or future-frame signal enters this space, which is what lets one head serve heterogeneous estimators.

D. Reset-and-Fusion Decision Head

The decision structure is inspired by CODD [1]. We do not claim reset/fusion or convex interpolation as novel in isolation.

The cache-grid pathway maps the 142-channel input to 24 channels through a 3×3 convolution, GroupNorm, SiLU, and a residual block. A coarse pathway expands to 48 channels at one-quarter resolution, followed by three 48-channel residual blocks.

A full-resolution reset branch predicts $r_t = \sigma(z_t^r)$, while the coarse fusion branch predicts $f_t = \uparrow_4 [\sigma(z_t^f)]$.

The effective temporal weight is $w_t = r_t f_t$.

Temporal evidence is only defined where the aligned memory is itself usable. We therefore define the per-pixel *support* as the conjunction of the current raw validity, the aligned temporal validity, and the warp support,

$$S_t = M_t^{\text{raw}} \wedge \widetilde{M}_t \wedge \mathcal{S}_t, \tag{3}$$

and confine the intervention to it. Final disparity is

$$\boxed{d_t^{\text{fused}} = \begin{cases} (1-w_t)d_t^{\text{raw}} + w_t\widetilde{m}_t, & S_t = 1, \\ d_t^{\text{raw}}, & S_t = 0. \end{cases}} \tag{4}$$

Because $w_t \in [0, 1]$, on the support

$$\min(d_t^{\text{raw}}, \widetilde{m}_t) \le d_t^{\text{fused}} \le \max(d_t^{\text{raw}}, \widetilde{m}_t), \tag{5}$$

and off the support the module is the exact identity on raw geometry.

The refiner therefore cannot synthesize arbitrary disparity outside the interval spanned by its two candidates.

Applying (3) to the output — rather than only computing it — is a load-bearing part of the contract, not a formality. The causal pull-warp uses zero padding, so an out-of-bounds sample returns $\widetilde{m}_t = 0$ *exactly*. An unmasked convex blend at such a pixel can therefore drive d_t^{fused} to zero, producing an invalid disparity where the frozen stereo estimator produced a valid one, and the bounded recurrence of Sec. III-E then carries that zero forward until the next raw re-anchor. Section V-I quantifies the effect: it is invisible on SCARED-C and dominates the external-domain metric-depth tail.

The exact trainable parameter count is 177,338, and masking changes no parameter, no threshold, and no training procedure.

Reset output bias is initialized to -2, while fusion is initialized with zero bias, producing conservative initial temporal intervention.

E. Bounded Corrected-State Recurrence

After refinement, $m_t = d_t^{\text{fused}}$.

This creates the feedback path

$$d_{t-1}^{\text{fused}} \to \widetilde{m}_t \to d_t^{\text{fused}} \to \widetilde{m}_{t+1}. \tag{6}$$

Continuous propagation can therefore transform previous alignment and fusion errors into future evidence.

We explicitly bound corrected-state lifetime. On a sequence boundary the first output is raw. After a re-anchor, previous raw disparity is used as state. Otherwise, the previous fused output is propagated.

The canonical policy permits four corrected-state uses before a raw re-anchor. Thus H denotes state lifetime rather than a temporal input window.

The method remains one-step causal: $\{I_{t-1}, I_t, m_{t-1}\} \to d_t^{\text{fused}}$.

F. Training Objective

For target disparity g_t let $e_t^{\text{raw}} = |d_t^{\text{raw}} - g_t|$, $e_t^{\text{mem}} = |\widetilde{m}_t - g_t|$ and $\Delta_t = e_t^{\text{mem}} - e_t^{\text{raw}}$, so $\Delta_t < 0$ marks pixels where aligned memory is the better candidate. Training support M is the intersection of GT coverage, current validity, aligned temporal validity and warp support.

The objective is $\mathcal{L} = \mathcal{L}_{disp} + \mathcal{L}_{reset} + \mathcal{L}_{fusion}$. The reconstruction term is $\mathcal{L}_{disp} = \mu_M[\text{Huber}_{\delta=1}(d_t^{\text{fused}} - g_t)]$. The two decision terms supervise the branches directly against Δ_t at native margins $\tau_r^n = 5\,\text{px}$ and $\tau_f^n = 1\,\text{px}$ (cache scale 0.703125 and 0.140625): writing $R^{\pm}$ and $F^{\pm}$

for the sets where Δ_t exceeds $\pm\tau_r^c$ and $\pm\tau_f^c$, and F^0 for $|\Delta_t| \le \tau_f^c$,

$$\mathcal{L}_{reset} = \mu_{R^+}[r_t] + \mu_{R^-}[1 - r_t], \quad (7)$$

$$\mathcal{L}_{fusion} = \mu_{F^+}[f_t] + \mu_{F^-}[1 - f_t] + 0.2\,\mu_{F^0}[|f_t - 0.5|]\,. \quad (8)$$

No explicit temporal-consistency objective is used: temporal behaviour is a consequence of the decision, never a training target.

IV. Experimental Protocol

A. Canonical Training

The canonical temporal refiner is trained on temporally curated SCARED-C data derived from SCARED [20].

Training uses frozen predictions from $\mathrm{S}^2\mathrm{M}^2$-S [15], RAFT-Stereo [12] and Stereo Anywhere [14]; CREStereo [13] and Fast-FoundationStereo [17] are excluded from temporal-head training, and no backbone identity is ever given to TETHER.

The canonical configuration trains on D1, D3 and D6 with D2 for development and D7 held out, over 8,616 frames giving 8,606 causal pairs from three seen backbones, using non-overlapping $H = 4$ clips, batch size 4, 12 epochs of AdamW at 2×10^{-4} with weight decay 10^{-4} and gradient clipping 1.0. The 177,338 trainable parameters are selected on development fused EPE.

B. External Evaluation

a) DRENDS.: DRENDS contains dynamic robotic-endoscopy sequences with stereo imagery, calibration, and metric geometry [24]. We report five curated recordings — `Vid10_Liver_Med`, `Vid11_Liver_High`, `Vid12_Pancreas_Ext`, `Vid13_Pancreas_Med` and `Vid14_Pancreas_High` — evaluated with three stereo estimators for 20,929 frames in total. The zero-shot claim applies only to the SCARED-trained canonical checkpoint: no DRENDS data enters training, checkpoint selection or threshold calibration.

Disparity reference is derived from metric ToF geometry through stereo calibration. It is therefore useful metric evidence but not independent stereo ground truth, and the ToF reference itself is temporally filtered.

b) D4D.: D4D contains deformable surgical stereo sequences and structured-light observations [23]. Because the current geometric scale contract remains unresolved, we use D4D only for no-reference temporal diagnostics.

c) SERV-CT.: SERV-CT provides calibrated stereo pairs and CT-derived reference geometry [21]. The evaluated pairs are non-consecutive, so temporal refinement is marked not applicable.

C. Unified Temporal-Refinement Evaluation

A central requirement is that evaluation support cannot depend on the prediction: $M_{\mathrm{eval}} = M_{\mathrm{GT}} \cap M_{\mathrm{protocol}}$.

Invalid predictions on valid support are penalized rather than silently removed.

The framework evaluates five complementary dimensions.

a) Spatial geometry.: For disparity we report EPE/MAE, median, RMSE, P90/P95/P99, maximum, invalid rate, and Bad-$1/3/5$. For metric depth we additionally report Bad-$2/5/10$ mm, AbsRel, SqRel, and $\delta_{1,2,3}$.

b) Temporal fidelity.: At horizons $k \in \{1, 2, 4, 8\}$ the framework defines grid-based temporal change error and, when correspondence is supplied, motion-compensated $\mathrm{TEPE}_{\mathrm{corr}}$, warped depth DTCE and pure prediction consistency (OPW). Estimated-flow correspondences are marked as flow-conditioned proxies. We report the motion-compensated family in the supplementary material: it is informative — on held-out D7 both $\mathrm{TEPE}_{\mathrm{corr}}$ and OPW improve in all $20/20$ backbone–horizon cases — but on development D2 the two disagree, which needs more space to treat honestly than a submission of this length allows.

c) Intervention harm and benefit.: For $\delta e = e^{\mathrm{fused}} - e^{\mathrm{raw}}$, we measure

$$H^+ = \mathbb{E}[\max(\delta e, 0)], \quad (9)$$

$$B^+ = \mathbb{E}[\max(-\delta e, 0)], \quad (10)$$

$$\mathrm{HUR} = P(\delta e > 0), \quad (11)$$

$$\mathrm{BUR} = P(\delta e < 0). \quad (12)$$

For each bad-pixel threshold τ we additionally report NewBad, RecoveredBad, bad-rate delta, and identity preservation.

d) Frame-level tails.: For each frame,

$$D_t = \frac{1}{|M_t|} \sum_{x \in M_t} (e_t^{\mathrm{fused}}(x) - e_t^{\mathrm{raw}}(x)). \quad (13)$$

We retain mean, median, P95, P99, worst degradation, and the fraction of degraded frames.

e) Selective risk.: When a risk score is available, the same framework supports risk–coverage analysis, AURC, oracle/excess AURC, AUROC, AUPRC, Brier, NLL, and ECE [19].

Primary aggregation is equal-sequence macro averaging. Pixel-weighted micro statistics are retained secondarily.

V. Results

A. Does the Learned Temporal Policy Matter?

We first isolate the contribution of the learned decision policy on D2. All variants use the same frozen stereo predictions, temporal alignment, support contract, and evaluation implementation. Only the temporal policy is changed.

The canonical learned policy obtains $\Delta_{\mathrm{TETHER}} = 0.276202 - 0.265028 = 0.011174$ px.

The best matched fixed fusion obtains only $\Delta_{\mathrm{fixed}} = 0.276202 - 0.274562 = 0.001640$ px.

Therefore, $\Delta_{\mathrm{TETHER}}/\Delta_{\mathrm{fixed}} \approx 6.8$: the learned decision recovers almost seven times the gain of the best fixed blend under an identical state machine.

Forward–backward confidence is the informative failure. It is one of TETHER's own input cues, yet using it directly as the fusion weight worsens EPE by 2.17%: motion reliability is useful evidence and a poor decision rule. Propagating aligned memory directly, or letting corrected recurrence run

TABLE I

SCARED-C D2 MATCHED TEMPORAL-POLICY CLOSURE. RAW EPE IS 0.276202. POSITIVE REDUCTION DENOTES IMPROVEMENT.

Method	EPE	Reduction
TETHER $H = 6$	**0.264656**	4.18%
TETHER $H = 4$	0.265028	4.05%
TETHER $H = 1$	0.269001	2.61%
Fixed $w = 0.5 \equiv$ EMA2	0.274562	0.59%
Fixed $w = 2/3 \equiv$ EMA3	0.275386	0.30%
Fixed $w = 0.1$	0.275684	0.19%
Warped raw previous	0.276298	-0.03%
FB confidence weight	0.282182	-2.17%
Warped recurrent $w = 1$	0.286035	-3.56%
Continuous recurrence	0.449349	-62.69%
Oracle raw/memory	0.242223	12.30%

unbounded, degrades accuracy as well. The improvement is therefore not explained by alignment, averaging or smoothing.

One entry deserves a caveat: under the implemented one-step update, EMA2 is algebraically identical to fixed $w = 0.5$, so their equal outputs are expected rather than two independent confirmations.

B. Why $H = 4$?

The lowest development EPE is at $H = 6$ (0.264656 versus 0.265028), a difference of 0.000372 px or 0.14%. Longer state lifetime buys that sliver at a measurable cost in intervention risk: from $H = 4$ to 6 to 8 the harmful-update rate rises (0.4048, 0.4053, 0.4068) while the beneficial rate falls (0.5074, 0.5065, 0.5047) and B^+/H^+ degrades from 2.53 to 2.30 to 2.04; the frame tail follows, with $P95(D_t)$ at 0.01715, 0.02037, 0.02441 and NewBad1 at 0.0467%, 0.0676%, 0.1282%. We therefore do not claim $H = 4$ minimises development EPE. It is the frozen risk–gain operating point, chosen before the held-out split was opened.

C. Why Recurrence Must Be Bounded

Continuous corrected-state recurrence moves mean EPE from $0.276202 \rightarrow 0.449349$, a 62.69% increase: writing corrected disparity back into memory without interruption accumulates alignment and fusion error. Periodic raw re-anchoring is therefore structural rather than an implementation detail, though this does not imply that every continuous recurrent architecture must fail.

D. Cross-Backbone Development Behaviour

Canonical $H = 4$ improves all five estimators in mean D2 EPE (Table II): 8.44% on StereoAnywhere, 3.37% on CREStereo, 2.87% on RAFT-Stereo, 2.85% on Fast-FoundationStereo, 2.17% on S^2M^2-S; seen backbones 4.50%, unseen 3.11%, with all six unseen-backbone sequence groups improving. The pooled 4.05% is the ratio of mean EPEs, not the mean of those percentages (3.94%, median 2.87%). It is fair to ask whether StereoAnywhere alone carries the result: excluding it the D2 mean falls to 2.82%, but on held-out D7 it does not, the mean excluding StereoAnywhere being 4.14% with RAFT-Stereo alone at 6.10%.

E. Held-Out Cross-Backbone Confirmation

We next evaluate the already frozen $H = 4$ policy on held-out D7.

All five backbones improve on D7, including both architectures unseen during temporal-head training.

The mean backbone EPE decreases from $0.4921 \rightarrow 0.4696$, or 4.57%.

The improvement is not restricted to the mean. Bad1 and disparity RMSE decrease for every backbone on both D2 and D7. P99 disparity error likewise decreases in all ten backbone–split cases, while maximum error decreases in nine of ten. The strongest tail reduction occurs for Stereo Anywhere on D2, where maximum disparity error decreases from 33.27 to 14.88 px. InvalidRate remains zero for both raw and refined outputs in all ten SCARED-C backbone–split evaluations. This is a property of SCARED-C rather than of the module in general: Section V-I shows that on external domains an unmasked output does introduce invalid predictions, which is precisely what the support contract of (4) prevents.

Thus the EPE gain is not obtained by trading mean accuracy for a worse aggregate bad-pixel rate or a broader SCARED-C error tail. This does not imply that no individual new bad pixels are introduced; those events are measured explicitly through NewBad and recovered-error statistics below.

This supports transfer of one common learned intervention policy across heterogeneous stereo estimators within the SCARED-C domain.

The joint unseen-backbone / unseen-domain cell is reported separately in Section V-H.

F. Seed Variance, and What It Reveals About Selection

We pre-registered a seed study before any seed result existed: the seed is the only field that varies, sequences are the resampling unit, seeds are averaged inside each resample, and selecting the best seed is prohibited.

Pooled over five backbones, the relative Bad1 reduction is $8.40 \pm 2.53\%$ on development D2 and $5.15 \pm 0.35\%$ on held-out D7; the EPE reduction is $3.55 \pm 0.45\%$ and $5.07 \pm 0.44\%$.

The held-out effect is stable: 5.15 ± 0.35 Bad1 and 5.07 ± 0.44 EPE, with paired sequence-level bootstrap intervals $[4.11, 8.02]$ and $[3.42, 5.80]$ and no resample in which refinement failed to help.

The development split is not, and our own protocol explains it. Checkpoints are selected on best D2 fused EPE, so seed 0 is by construction the run that looked best on D2 — and on D2 the seeds fall monotonically, $10.78 \rightarrow 8.67 \rightarrow 5.75$, a 5.03 point spread. On D7 that ordering *inverts*: seed 0 is the weakest of the three on both metrics and the spread collapses to 0.69 points. Selection optimism sits where it should, in the split used to select, and leaves the held-out numbers untouched. Table II keeps seed 0 for comparability with the ablations, which use that checkpoint.

TABLE II

CROSS-BACKBONE EVALUATION ON SCARED-C. THE SAME FROZEN TETHER CHECKPOINT IS USED IN EVERY ROW. EPE, BAD1, AND RMSE USE THE PRIMARY MACRO-SEQUENCE AGGREGATION. LOWER IS BETTER.

		D2 development			D7 held out		
Backbone	Status	EPE raw→TETHER	Bad1 raw→TETHER	RMSE raw→TETHER	EPE raw→TETHER	Bad1 raw→TETHER	RMSE raw→TETHER
S^2M^2-S	seen	.2704→**.2645**	.01394→**.01291**	.3892→**.3806**	.5176→**.5010**	.06440→**.06076**	1.4515→**1.3991**
RAFT-Stereo	seen	.2691→**.2614**	.01603→**.01492**	.4009→**.3896**	.4671→**.4397**	.06085→**.05845**	1.3324→**1.2290**
Stereo Anywhere	seen	.3022→**.2767**	.02183→**.01754**	.7208→**.4785**	.4774→**.4449**	.06227→**.05828**	1.2869→**1.1895**
CREStereo	unseen	.2668→**.2578**	.01568→**.01439**	.4601→**.4299**	.5165→**.4957**	.06354→**.06108**	1.3757→**1.3120**
Fast-FoundationStereo	unseen	.2725→**.2647**	.01604→**.01477**	.4022→**.3905**	.4819→**.4668**	.05919→**.05684**	1.3535→**1.3076**

G. Refinement as an Intervention

Beneficial updates consistently outweigh the threshold crossings the module introduces. Table III reports the beneficial-to-harmful error-magnitude ratio B^+/H^+ and the ratio of recovered to newly introduced Bad1 pixels.

TABLE III

CANONICAL TETHER INTERVENTION QUALITY. ALL RATIOS EXCEED ONE; HIGHER IS BETTER. "REC./NEW" IS RECOVEREDBAD1/NEWBAD1.

	D2		D7	
Backbone	B^+/H^+	Rec./New	B^+/H^+	Rec./New
S^2M^2-S	1.76	3.1×	2.10	6.0×
RAFT-Stereo	2.05	2.9×	2.85	3.2×
Stereo Anywhere	4.59	10.4×	3.30	5.9×
CREStereo	2.23	4.3×	2.40	4.6×
Fast-FoundationStereo	2.11	4.0×	2.10	4.8×

Across both splits and all five backbones, TETHER recovers between $2.9\times$ and $10.4\times$ as many Bad1 pixels as it newly introduces. The corresponding identity-preservation score exceeds 99.87% in every case. This directly supports the refinement-centric interpretation: the method changes only a small subset of threshold status while recovering substantially more errors than it creates.

H. External Zero-Shot Geometry on DRENDS

We evaluate the frozen canonical checkpoint on five DRENDS recordings with three stereo estimators, two of which were excluded from temporal-head training, for a total of 20,929 frames. No component is adapted, retrained, or tuned for this domain.

Across the 15 recording–backbone combinations, disparity EPE, Bad3, metric depth MAE and metric depth RMSE all improve in *every* cell (60/60), with mean changes of -5.00%, -17.35%, -3.68% and -3.21% respectively.

Two observations matter more than the aggregate. First, the per-recording pattern is set by the *recording* and not by the estimator: on Vid10 the EPE change is -5.7%, -5.2% and -5.0% for the seen and the two unseen backbones respectively, and the same alignment holds on every other recording. The residual variance is therefore a property of the data, which is stronger evidence for backbone independence than any pooled mean. Second, both unseen estimators improve on all five recordings, so the joint unseen-backbone / unseen-domain cell is positive rather than merely unevaluated.

The transfer matrix is consequently complete: seen-backbone / seen-domain and unseen-backbone / seen-domain on SCARED-C (Table II), and seen-backbone / OOD-domain and unseen-backbone / OOD-domain on DRENDS (Table IV). This establishes zero-shot geometric transfer in disparity space under a ToF-derived metric reference; it does not establish universal cross-domain robustness, and DRENDS remains a non-independent reference.

I. The Support Contract Is Load-Bearing

The results above use the support-confined output of (4). Because the ablation is a single line, it isolates the contract cleanly: the checkpoint, the head, the evidence, the policy and the reset schedule are identical, and only the masking of the output differs.

Without the mask, zero-padded out-of-bounds memory enters the convex blend as $\widetilde{m}_t = 0$, the fused disparity collapses at those pixels and bounded recurrence propagates the collapse to the next re-anchor. The offenders are rare and bursty — 300 of 38,828,160 on Vid10_Liver_Med, 265 of 38,439,360 on Vid11_Liver_High, in runs of at most H frames — and the aligned memory of every one is exactly 0.0. Because an invalid prediction on valid support is penalised rather than discarded, those few hundred pixels are the entire source of the depth tail previously attributed to the external domain. Confining the output removes it and improves disparity too, since the zero no longer enters memory.

On SCARED-C the same ablation is nearly inert: D2 is unchanged at 4.05% and D7 falls from 4.64% to 4.57%, the largest single change being RAFT-Stereo ($6.10\% \to 5.87\%$). The contract is therefore not free in domain; it costs a little in-domain gain and buys well-posed behaviour out of domain. We report the confined variant because a module that declares a support must respect it.

J. No-Reference Temporal Behaviour on D4D

Because the D4D metric-geometry contract remains unresolved, we report only prediction-space temporal inconsistency, which falls by 51.9% for RAFT-Stereo and 54.5% for StereoAnywhere. That fact on its own is worth very little, and we quantify why with two degenerate controls that only reproduce history. Copy-previous, which emits the previous disparity verbatim without even aligning it, removes 97.3% of the inconsistency; warped-previous, which emits the motion-compensated previous disparity, scores *exactly*

TABLE IV

ZERO-SHOT DRENDS TRANSFER, RELATIVE CHANGE AFTER REFINEMENT (NEGATIVE IS BETTER). THE REFERENCE DISPARITY IS DERIVED FROM TEMPORALLY FILTERED TOF GEOMETRY AND IS METRIC EVIDENCE, NOT INDEPENDENT STEREO GROUND TRUTH. THE TWO RIGHTMOST ESTIMATORS WERE EXCLUDED FROM TEMPORAL-HEAD TRAINING, SO THEIR COLUMNS ARE THE JOINT UNSEEN-BACKBONE / UNSEEN-DOMAIN SETTING.

	RAFT-Stereo (seen)		CREStereo (unseen)		Fast-FoundationStereo (unseen)	
Recording	EPE	Bad3	EPE	Bad3	EPE	Bad3
Vid10_Liver_Med	-5.7%	-11.3%	-5.2%	-8.7%	-5.0%	-7.3%
Vid11_Liver_High	-3.4%	-21.9%	-3.7%	-22.8%	-3.2%	-20.4%
Vid12_Pancreas_Ext	-8.4%	-22.7%	-6.3%	-18.5%	-5.3%	-10.3%
Vid13_Pancreas_Med	-4.3%	-14.2%	-5.2%	-10.4%	-4.3%	-10.9%
Vid14_Pancreas_High	-4.7%	-27.1%	-5.6%	-26.5%	-4.7%	-27.4%

TABLE V

EFFECT OF APPLYING THE SUPPORT MASK OF (3) TO THE MODULE OUTPUT, OVER FIVE DRENDS RECORDINGS AND THREE BACKBONES. "REGRESSES" COUNTS RECORDING–BACKBONE CELLS THAT GET WORSE.

Quantity	Unmasked output	Support-confined
Invalid refined pixels	up to 1.3×10^{-5}	**0**
Depth RMSE	regresses $14/14$, $+110.4\%$	**improves** $15/15$, -3.2%
Depth MAE	regresses $10/14$, $+4.2\%$	**improves** $15/15$, -3.7%
Disparity EPE	-4.30%	-5.00%
Disparity Bad3	-14.06%	-17.35%

zero by construction, since a policy whose output is the warped previous frame is trivially consistent with the warped previous frame. TETHER removes 53.9%.

The ordering is inverted with respect to geometric quality: the two policies that discard current stereo evidence outrank the one that uses it. We therefore treat D4D no-reference consistency as necessary but far from sufficient, make no D4D geometric claim, and report the controls beside the metric so no reduction figure can be read as accuracy. The full table is in the supplementary material.

VI. ANALYSIS

A. What Makes TETHER Work, and Why So Few Parameters Suffice

The closure isolates four requirements. Geometric alignment is necessary but not sufficient: the aligned memory carries a 12.30% oracle ceiling, yet using it indiscriminately fails, since warped raw memory yields no meaningful gain and both direct FB-confidence weighting and full recurrent replacement worsen EPE. Under identical evidence and support the learned policy obtains almost seven times the gain of the best fixed blend, and the recurrent state must stay bounded because continuous propagation accumulates error and collapses. The mechanism is motion alignment plus learned graded intervention plus bounded recurrent state, not generic smoothing.

This also explains the parameter count. The head never solves stereo matching: it receives two already meaningful candidates, d_t^{raw} and $\widetilde{m}_t$, plus evidence describing their disagreement, motion reliability, validity, matching consistency and appearance, so its problem is closer to judging whether temporal evidence is useful than to predicting disparity from RGB. Convex fusion further restricts the action space to a position between two candidate geometries. A decision problem that constrained is a plausible reason why one small head transfers to architectures absent from its training.

B. The Oracle Gap Is the Next Problem

TETHER captures only 32.9% of the raw-versus-memory oracle gain, and 30.1% of the continuous oracle that its own convex action space could in principle reach; only 6.1% of D2 pixels have the ground truth bracketed by the two candidates, so the two ceilings nearly coincide. Useful evidence therefore already exists, and the open problem is discriminating helpful from harmful intervention.

C. The Fusion Weight Is Not an Uncertainty

Recovering the effective weight exactly from the convex form, $w_t = (d_t^{\text{fused}} - d_t^{\text{raw}})/(\widetilde{m}_t - d_t^{\text{raw}})$, and scoring it as a predictor of harmful intervention gives AUROC 0.366–0.382 on D2 but 0.496–0.506 on held-out D7, i.e. chance; the raw–memory disagreement and the update magnitude behave the same way. The reliability curve inverts between splits — for RAFT-Stereo the harm rate falls with w_t on D2 ($0.550 \to 0.324$) and rises on D7 ($0.260 \to 0.450$) — so w_t is not a weak risk score but a sign-unstable one, capturing only 3.5% and 6.1% of the achievable selective gain. It is an intervention magnitude, not an uncertainty, and closing the oracle gap needs evidence the head does not receive.

D. Evaluation Matters

The framework exposes behaviour a mean would hide: on DRENDS mean and percentile geometry improve while RMSE worsens; the augmented model raises both beneficial and harmful intervention; longer horizons slightly improve mean D2 EPE while degrading B^+/H^+, NewBad1 and the frame tails. Temporal refinement is better read as an *intervention* than as a final estimator.

One reported quantity failed an audit and is withdrawn. FrameDegradation's CatastrophicFraction equalled PositiveFraction on every inspected row — definitional, not a wiring fault: the two are $\Pr(D_t > \tau_{\text{cat}})$ and $\Pr(D_t > 0)$, and τ_{cat} defaults to 0, which no evaluator overrides. We report the degraded-frame fraction $\Pr(D_t > 0)$ and drop the duplicate column rather than keep a name implying a severity threshold it does not carry.

VII. Discussion and Limitations

a) What the splits support.: The matched-baseline and horizon closure is development work on D2: it supports mechanism analysis and the pre-frozen $H = 4$ operating point, not confirmation. The frozen checkpoint is evaluated on D7 without changing the policy. Transfer is demonstrated along the backbone axis, the domain axis and their combination, but a single external domain does not establish universal cross-domain robustness, and DRENDS supplies a temporally filtered ToF reference rather than independent stereo ground truth. Per-backbone D7 margins come from one canonical seed and the three-seed study bounds only the pooled effect, so individual backbone rows are not separated from one another.

b) What the metrics cannot say.: No-reference temporal consistency is gameable — replaying warped history scores exactly zero — so we make no D4D geometric claim. Pull-warp support detects out-of-frame sampling but cannot model internal disocclusion, so newly visible anatomy may inherit stale geometry despite plausible motion consistency; this remains a plausible source of residual tails.

c) Cost and scope.: The head is compact but the deployed system also runs frozen stereo and two SEA-RAFT passes, giving $33.60\,$ms of temporal overhead at $124.2\,$MB peak VRAM on an H100, hence 11.0 down to $1.4\,$FPS end-to-end depending on the estimator: *no real-time claim is made*. The experiments evaluate perception geometry and temporal behaviour, not downstream robotic control, and demonstrate neither clinical safety nor patient benefit.

VIII. Conclusion

TETHER attaches one causal temporal module, trained once, to frozen and heterogeneous stereo estimators without touching their internals or using future frames. Its gain is not temporal averaging: on development D2 it reduces EPE by 4.05% against 0.59% for the best matched fixed blend under an identical state machine, while direct flow-confidence weighting and unbounded recurrence degrade accuracy, the latter by 62.69%. A frozen $H = 4$ checkpoint improves all five estimators on held-out D7 in EPE, Bad1 and RMSE simultaneously, including two excluded from training, recovers 2.9–$10.4\times$ more Bad1 pixels than it introduces, and holds at $5.15 \pm 0.35\%$ Bad1 across three independently trained seeds. Zero-shot transfer to five DRENDS recordings improves every one of 60 recording–backbone–metric cells.

Three results bound what may be claimed. The no-reference consistency metric is won outright by a policy that merely replays warped history, so smoothness cannot stand in for correctness. The fusion weight inverts its relationship with harm between development and held-out data, so it is an intervention magnitude and not an uncertainty. And development gains carry selection optimism that held-out gains do not: across seeds the D2 effect spans 5.03 points and the D7 effect 0.69. The oracle over the module's own action space remains far from reached, placing the open problem in deciding *where* to intervene rather than in obtaining temporal evidence.

Finally, the support a module declares must also be applied to what it emits. Confining the fused output to it costs 0.07 percentage points in domain and removes the entire external-domain metric-depth tail.

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